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Real-World Performance of an AI System for Diabetic Retinopathy Screening

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Abstract

Diabetic retinopathy (DR) is a leading cause of preventable blindness, and the growing global burden of diabetes is placing increasing pressure on ophthalmic services. Artificial intelligence (AI)-based retinal image analysis offers a promising strategy to scale up DR screening while reducing reliance on specialist graders. We assessed the performance of an AI-based DR screening system implemented in a real-world endocrinology clinic at the Erasmus Hospital, Belgium. Adult patients with diabetes underwent non-mydriatic fundus photography, and images were analyzed by the AI system for referable DR and diabetic macular edema. All images were independently graded by a retinal specialist using the Early Treatment Diabetic Retinopathy Study (ETDRS) classification as the reference standard. Of 405 patients screened, 353 (86.7%) were included in the primary analysis. The AI system achieved an area under the curve of 96.5%, sensitivity of 88.9%, specificity of 98.7%, and high predictive values for referable DR detection. Subgroup analyses showed consistently high accuracy across demographic and clinical strata. Multivariate analysis identified higher HbA1c at diagnosis and longer diabetes duration as significant predictors of referable DR for both AI and human grading. These findings support the robustness, generalizability and operational feasibility of this AI system for DR screening in routine clinical care.

Keywords: diabetic retinopathy; artificial intelligence; deep learning; screening; teleophthalmology; real-world study

Introduction

Diabetes mellitus is characterized by chronically elevated blood glucose levels that progressively damage the inner lining of blood vessels and lead to long-term microvascular complications. Diabetic retinopathy (DR) is the most common microvascular complication of diabetes and a leading cause of vision loss and preventable blindness in working-age adults worldwide. The global prevalence of diabetes is rising at an alarming rate: 589 million adults were living with diabetes in 2024, with projections reaching 853 million by 2050 [1]. As the number of individuals with diabetes increases, the cost and organization of eye care will become even more critical [2].

Early detection through regular screening is essential to prevent vision loss, but the growing demand for screening challenges an already under-resourced ophthalmic workforce [3]. In some countries, such as the United Kingdom and Iceland, nationwide DR screening programs have successfully reduced DR-related blindness to the point where it is no longer the leading cause of blindness in working-age adults [4,5]. In Belgium, however, DR screening remains largely opportunistic and fragmented, and a more structured approach could reduce inefficiencies and delays in care.

Traditional DR screening is based on dilated fundus examination by ophthalmologists, which is labor-intensive, resource-demanding and costly for health systems [6,7]. Task differentiation, whereby image acquisition and grading are delegated to trained non-physician staff, can streamline the process and allow ophthalmologists to focus on treatment. In parallel, advances in artificial intelligence (AI), particularly deep learning based on convolutional neural networks, have enabled automated image recognition in highly image-driven specialties such as ophthalmology, dermatology and radiology [8]. In ophthalmology, AI-based algorithms for DR detection have demonstrated high diagnostic performance on retinal fundus photographs [9,10].

Several AI systems have since been developed for automated, real-time DR detection and have generally shown accuracy comparable to or exceeding that of human graders [11]. However, most studies focus on model development and validation under controlled conditions, whereas real-world implementation studies remain limited and are mainly restricted to highly structured national screening programmes, such as in the UK [10,12]. In clinical practice, variable disease prevalence, heterogeneous patient populations and inconsistent image quality can affect algorithm performance. Real-world evaluations are therefore critical to understand the robustness, usability and clinical impact of AI-based DR screening [13].

The primary objective of this study was to evaluate the accuracy of a recently implemented AI-based screening system for referable DR in a Belgian population with diverse ethnic backgrounds at Erasmus Hospital, Brussels University Hospital. Secondary objectives were to assess the generalizability of the system across subgroups and to investigate its ability to identify systemic risk factors associated with referable DR, hypothesizing that AI performance would be at least equivalent to that of human graders.

Methods

Ethics

The study was approved by the Erasmus Hospital HUB Ethics Committee (reference P2023/464). All participants provided informed consent, and all procedures were conducted in accordance with the Declaration of Helsinki and relevant institutional guidelines and regulations.

Patients

All adult patients (≥ 18 years) with a diagnosis of diabetes attending consultations at the Erasmus Medical Center were invited to participate between 10 January and 20 April 2024. Recruitment took place immediately following their usual diabetes consultation.

AI-assisted DR screening system

The AI-based screening model was developed by MONA.health, a Belgian healthcare AI company supported by KU Leuven and the Health Unit of VITO. The system autonomously screens for referable DR and diabetic macular edema and its development and validation details have been reported previously [14]. Briefly, the system is based on an ensemble of convolutional neural networks that first perform automated image quality assessment, followed by binary detection of referable DR and diabetic macular edema. Multiple convolutional neural network architectures are combined to generate a patient-level referral decision using a fixed operating threshold targeting high sensitivity, as previously described. The algorithm was trained and tested on a large multicenter private dataset from a US-based screening network including individuals of African descent, Asian, Caucasian, Latin American and unspecified ethnicity, as well as a publicly available dataset comprising mainly US and French images commonly used for benchmarking DR detection algorithms.

Fundus image acquisition

For each participant, a single 45-degree, non-dilated, color fundus photograph of each eye was acquired using a Crystalvue camera, centered between the optic disc and macula. The MONA.health software independently performs an automated image quality assessment, and requires at least one image of sufficient quality per eye to generate a report on the patient's referral status for DR and diabetic macular edema. If the maximum predicted value for at least one eye exceeded the fixed thresholds (1.371 for DR and 0.38 for diabetic macular edema, corresponding to an operating point targeting 90% sensitivity), the patient was flagged for referral for one or both conditions. Referable DR was defined as moderate non-proliferative DR or worse, diabetic macular edema, or an ungradable image, based on the Early Treatment Diabetic Retinopathy Study (ETDRS) classification [14,15].

Screening protocol

Prior to patient recruitment, nurses and a medical student (LB) received standardized one-to-one training by the same trainer, covering retinal image acquisition, troubleshooting for poor image quality, automated grading using the AI system and communication of results to patients. Training and supervision were provided until staff were comfortable performing data collection and imaging independently.

Following image acquisition, retinal images were uploaded to the AI system, which generated an automated referral status report. Results were orally communicated to patients and incorporated into their medical records at the time of screening. Patients who tested positive for referable DR and/or diabetic macular edema and were previously undiagnosed, or who had insufficient image quality, were invited to consult an ophthalmologist. The Ophthalmology Department at Erasmus Hospital contacted these patients for follow-up if needed.

Reference standard grading

To determine the accuracy of the AI screening model, all retinal images were manually graded for referable DR by a retinal specialist (EM) at Erasmus Hospital using the ETDRS classification. As the reference standard, referable DR was defined as moderate non-proliferative DR or vision-threatening stages (severe non-proliferative and proliferative DR) in at least one eye. When grading differed between eyes, the presence of referable DR in at least one eye was considered sufficient to classify the patient as having referable DR, in accordance with ETDRS criteria. The AI system applies the same clinical definition of referable DR but provides a binary referable/non-referable output without further severity stratification. Human grading was performed exclusively for reference standard evaluation and was not part of the AI screening workflow. Patients identified as false negatives (no DR reported by AI in either eye but DR detected on manual grading) were contacted by telephone and advised to attend an ophthalmologic consultation.

Statistical analysis

All analyses were performed using RStudio (version 4.3.3). Patients with missing reference standard grading due to missing images in at least one eye (small pupil, poor fixation, technical error, upload error) or ungradable images according to either the AI system or manual grading were excluded from the primary analysis.

Sensitivity, specificity, positive predictive value and negative predictive value for referable DR detection were estimated with 95% exact binomial confidence intervals. AUC and corresponding confidence intervals were estimated using a bootstrap technique with 1,000 replicates. All diagnostic accuracy analyses were performed at the patient level. The highest predicted probability across both eyes was used for classification and AUC estimation, as referral is triggered when referable DR is detected in at least one eye.

The same diagnostic accuracy analysis was performed in predefined subgroups based on age, sex, ethnicity, diabetes type, HbA1c at diagnosis, diabetes duration, BMI and image quality. Image quality was categorized as excellent, good, adequate or “dirty lens”; for subgroup analysis, “excellent” corresponded to excellent quality in both eyes, “good” to good quality in at least one eye, and “adequate/dirty lens” to adequate or dirty lens quality in both eyes (Figure 3). Comparisons of AUCs between subgroups were performed using the DeLong test.

A sensitivity analysis including all patients was undertaken to explore the potential impact of excluding individuals with missing reference standard diagnosis. In the worst-case scenario, all missing/unreadable images were treated as false negatives for referable DR; in the best-case scenario, all missing/unreadable images with referable features were treated as true positives and the remaining as true negatives.

Two multivariate logistic regression models were constructed to assess the association between systemic risk factors and referable DR, using AI-based and human grader-based classifications, respectively. Candidate variables were selected using a bootstrap procedure ($n = 500$), retaining only those that were significantly associated with referable DR in at least 50% of resamples. Model adequacy followed the rule of at least 10 events per explanatory variable [16]. Associations were summarized using odds ratios with 95% confidence intervals and a significance level of 5%.

Use of large language models

Parts of the text (grammar and style) were revised with the assistance of a large language model (ChatGPT, OpenAI). All scientific content, data interpretation and conclusions were determined and verified by the authors, who retain full responsibility for the work.

Results

Study population

A total of 405 patients with diabetes were invited to participate in the study. Of these, 353 (87.2%) were included in the primary analysis, while 52 (12.8%) were excluded due to missing or ungradable images. Reasons for exclusion included insufficient image quality ($n = 8$; 15.4%), missing images related to small pupils ($n = 14$; 26.9%), poor fixation ($n = 1$; 1.9%), technical issues such as incorrect positioning or duplicate images ($n = 8$; 15.4%), and images considered ungradable by the human grader ($n = 21$; 40.4%).

Among the 353 included patients, 192 (54.4%) were male. One hundred sixty-eight (47.6%) had type 2 diabetes, 113 (32.0%) had type 1 diabetes, and 42 (11.9%) had diabetes of unknown etiology. In the excluded group, 29 (55.8%) were male; 34 (65.4%) had type 2 diabetes, 7 (13.5%) had type 1 diabetes, and 9 (17.3%) had diabetes of unknown etiology. The median age of included patients was 56 years (range 19–89), and the median duration of diabetes was 13 years (0–56). Excluded patients were significantly older, with a median age of 68 years (25–88), and had a longer duration of diabetes (median 15 years; 1–45).

A majority of included and excluded patients, 213 (60.4%) and 28 (53.8%) respectively, reported that more than 12 months had elapsed since their last eye examination. Patient characteristics are summarized in Table I.

Referable DR was identified by the human grader in 54 patients (15.3%) and by the AI system in 59 patients (15.8%). A total of 31 discrepancies between AI and human grader diagnoses were observed: 6 false negatives, 4 false positives, and 21 cases that were ungradable according to human grader because of image quality issues. Among the ungradable cases, 13 were classified as non-referable and 8 as referable by the AI system (Table II).

A flow chart of the screening procedure and patient follow-up is shown in Figure 1. Examples of AI-generated grading reports are presented in Figure 2, and examples of images across different quality categories are shown in Figure 3.

Diagnostic accuracy and subgroup analysis

In the primary analysis, the prevalence of referable DR was 15.3% based on the human grader. The AI system demonstrated excellent diagnostic performance for the detection of referable DR. The area under the curve (AUC) was 96.5% (95% CI: 91.8–99.7), sensitivity was 88.9% (77.4–95.8), and specificity was 98.7% (96.7–99.7). The positive predictive value was 92.3% (81.4–97.9), and the negative predictive value was 98.0% (95.7–99.3). These values indicate strong agreement between the AI system and human grader for classifying patients as referable or non-referable (Table III). Among the referable cases identified by the human grader, 3 patients had vision-threatening DR, all of whom were correctly classified as referable by the AI system.

To assess the impact of excluding patients with missing or ungradable reference standard grading, a sensitivity analysis was performed on all 405 patients. In a worst-case scenario, all ungradable and non referable cases were assumed to be false negatives, leading to a substantial reduction in estimated sensitivity to 48.5%. In a best-case scenario, all ungradable and referable cases were treated as true positives and all ungradable and non-referable cases as true negatives, resulting in an estimated sensitivity of 90.2%. These analyses suggest that missing data may influence sensitivity estimates but do not fundamentally alter the high overall performance of the AI system.

Subgroup analyses were conducted according to age, sex, ethnicity, diabetes type, HbA1c at diagnosis, duration of diabetes, body mass index (BMI) and image quality (Table IV). The AI system maintained high AUC values across subgroups, with no statistically significant differences between groups. Diagnostic accuracy was particularly high in younger adults (19–52 years), women, European Caucasians, patients with type 1 diabetes and those with excellent image quality, although performance remained robust across all categories.

Multivariate analysis

Multivariate logistic regression models were constructed to evaluate systemic predictors of referable DR for both AI-based and human grader-

based classifications. Elevated HbA1c at diabetes diagnosis and longer duration of diabetes emerged as significant predictors of referable DR in both models. These two factors are well-established determinants of DR progression and were confirmed as strong risk markers in our cohort. Other variables, such as BMI and smoking status, were not significantly associated with referable DR, possibly due to the relatively low prevalence of referable DR and limited statistical power for these factors. Interestingly, other ethnicities (sub-Saharan African and Asian) were associated with an increased risk of referable DR in the AI model but not in the human grader model (Table V).

Patient referral and follow-up

Overall, 83 patients (23.5%) received a referral recommendation from the human graders for newly identified ocular findings. Among these, 36 (43.3%) were referred for retinal vascular tortuosity, 21 (25.3%) for asymmetric or atrophic optic discs, 14 (16.9%) for suspected age-related macular degeneration, 5 (6.0%) for optic disc hemorrhage or retinal lesions requiring further investigation, 5 (6.0%) for choroidal naevi and 4 (4.8%) for epiretinal membranes. Of all referred cases, 13 (15.7%) had referable DR. These results highlight the added value of fundus photography for detecting non-DR ocular pathologies that also warrant specialist assessment.

Discussion

In this real-world evaluation of an AI-based DR screening system, we observed high diagnostic accuracy, indicating robust performance in a routine clinical setting. Despite a smaller sample size compared with pivotal trials, the AI model exceeded FDA performance thresholds for DR screening systems (sensitivity >85% and specificity >82.5%) [17,18]. The AUC of 96.5%, coupled with high sensitivity and specificity, supports its suitability for identifying patients with referable DR in a heterogeneous diabetes population.

The MONA.health model was trained on a large US-based screening dataset including a majority of Latin American and unspecified ethnicities, as well as publicly available benchmarking data [14]. In our study, conducted in a predominantly European and North African Caucasian population, the AI system maintained excellent performance. This suggests promising generalizability beyond the original training distribution and supports the use of this system in diverse clinical environments. Our findings are supported by recent evidence from the large North East London screening program evaluating 8 CE-marked automated retinal image analysis systems, including MONA.health. In this study, MONA.health demonstrated high sensitivity for moderate-to-severe and proliferative DR across age, sex and ethnicity subgroups [12], in keeping with our findings. As DR screening pathways evolve, models capable of maintaining high accuracy across populations will be crucial.

Our findings are consistent with previous work showing that AI algorithms

for DR detection can perform at least as well in real-world conditions [12,13,19]. However, clinical implementation introduces variability in image acquisition, operator expertise and patient characteristics. In our cohort, excluded patients were significantly older and had a longer duration of diabetes, both of which are associated with more advanced ocular media changes, such as cataracts and vitreous floaters [14,20]. These factors may compromise image quality, particularly in non-mydriatic settings, leading to ungradable images and reduced screening coverage in the highest-risk individuals.

In routine clinical practice, however, ungradable images are systematically referred for ophthalmologic evaluation, as they may reflect ocular media opacities or advanced retinal disease. This real-world management pathway is therefore better reflected by the best-case scenario considered in our sensitivity analysis. Enhanced imaging protocols and targeted staff training may help to mitigate these limitations.

The majority of people with diabetes have either no DR or only mild DR and are therefore at low risk of imminent vision loss. In such low-risk populations, variation in human grader performance has limited impact on clinical outcomes, as management typically involves routine follow-up at one- to two-year intervals [21]. However, the cumulative time spent by clinicians on grading large volumes of non-referable images is considerable and reduces the capacity available for evaluating high-risk patients and managing active disease [22]. Our results demonstrate that an AI-based system can accurately triage patients, substantially reducing the human grading burden and allowing ophthalmologists to concentrate on complex or treatment-requiring cases. In line with our findings, previous real-world implementation studies have shown that automated retinal image analysis systems can substantially reduce the volume of images requiring human grading while maintaining high sensitivity for sight-threatening disease [12]. This aligns with the broader role of AI in automating repetitive tasks, performing them faster and with consistent quality [23].

In multivariate analysis, high HbA1c at diagnosis and longer diabetes duration were strongly associated with referable DR for both AI and human grader, confirming their role as major risk factors for DR progression [20]. Other factors, including BMI and smoking status, were not significantly associated with referable DR in this cohort, likely due to the modest number of referable DR events. These findings support the use of systemic risk profiling in conjunction with AI-based image analysis to prioritize patients for closer monitoring.

An important observation in our study was that 23.5% of patients were referred for newly discovered ocular findings, yet only 15.7% of these referrals were due to DR. This underscores the potential of fundus photography to uncover a broader spectrum of eye diseases, including retinal vascular anomalies, optic disc pathology, age-related macular degeneration and epiretinal membranes. In this context, while fundus photography enabled the identification of various non-diabetic ocular findings requiring specialist referral, the AI system was specifically designed

for DR and diabetic macular edema detection and does not aim to identify other ocular pathologies, underscoring its role as a triage rather than a comprehensive diagnostic tool. Future research should explore AI models capable of multi-disease detection (e.g. DR, glaucoma, Age-related macular degeneration, hypertensive retinopathy, high myopia and cataract) to avoid overlooking clinically significant non-DR pathologies [24].

This study has several strengths. To our knowledge, it is the first to evaluate the implementation of an AI-assisted DR screening software in real-world settings in Belgium. The system demonstrated clear time benefits in clinical workflow and maintained high diagnostic performance across diverse ethnic groups. Although time metrics were not formally recorded, real-world observations suggest that the AI-assisted screening workflow was efficient. The complete process, including patient installation, image acquisition, automated analysis and communication of results, required around 10 minutes per patient, while specialist review of retinal images required around 1 minute per case. In comparison, ophthalmology consultations generally require substantially more time, often 30-45 minutes, due to waiting times, pupil dilation and comprehensive examination. The concordance between AI and human graders in identifying patients at highest risk of DR, based on HbA1c and duration of diabetes, supports its clinical relevance for risk stratification.

Several limitations should also be acknowledged. First, the sample size is modest compared with larger multicenter studies [12,22,25], which may limit the precision of subgroup analyses and the generalizability of some secondary findings. Second, the AI system is cloud-based and requires stable internet connectivity, whereas offline systems may offer advantages in remote or resource-limited settings [26]. Third, we did not use the gold-standard seven-field stereoscopic retinal photography, which may have introduced some variability in referable DR classification. Fourth, the study did not assess patient adherence to referral recommendations or subsequent treatment, leaving the impact of AI-assisted screening on long-term visual outcomes uncertain. Acceptability of the AI-based screening tool among patients and clinicians was not formally assessed. However, refusal to participate was rare (<1%) and limited to logistical reasons. The system was readily integrated into routine diabetes consultations and ophthalmology referral pathways, suggesting good real-world acceptability. Beyond diagnostic accuracy, the real-world integration of AI-based screening systems is crucial. In this study, the AI system was implemented as a semi-autonomous tool integrated into routine care, providing automated image analysis and referral recommendations under medical supervision. Potential cost savings are expected to arise mainly from reduced specialist grading workload rather than full automation. Finally, we did not perform a formal cost-effectiveness analysis, which will be essential to inform policy decisions and support large-scale implementation [27,28].

In conclusion, our evaluation of an AI-based DR screening model in a real-world hospital setting demonstrates high accuracy, robustness and operational feasibility, with performance exceeding regulatory benchmarks for sensitivity and specificity. The system offers clear time savings and

appears generalizable across ethnically diverse populations. Further optimization of imaging protocols, long-term follow-up of referral pathways and health-economic analyses will be crucial to fully realize the benefits of AI-assisted DR screening and to guide its integration into national screening strategies.

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Author contributions

L.B., E.M. and M.C. designed the study. L.B. collected the data, performed the analyses and drafted the manuscript.

E.M. performed the reference grading and contributed to the methodological design.

M.L., M.C., L.C. and A.B. referred patients for the screening.

E.M. and M.C. supervised the study.

Data availability

The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request. Access is restricted due to GDPR and institutional data-protection policies.

Competing interests

The authors declare no competing interests

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Figure legends

Fig 1. Flow chart of screening procedures and follow-up of patients.

DR=diabetic retinopathy. NR= non referable. RDR=referable diabetic retinopathy

Fig 2. Examples of grading reports generated by AI-based screening system, including DR and diabetic maculopathy referral status.

Details of the development and validation of this AI system have been previously described [14].

Fig 3. Examples of images of dirty lens and insufficient, adequate, good, excellent quality.

Images collected during the present study using the MONA color fundus machine.

Tables

Characteristic	Included n=353	Excluded n=52	Total n=405	p value
<i>Age</i>				1.1×10^{-10}
Median (years, range)	56 (19-89)	68 (25-88)	-	
Mean (years)	54.80	69.15	-	
<i>Sex (n, %)</i>				0.97
Male	192 (54.4%)	29 (55.8%)	221 (54.6%)	
Female	161 (45.6%)	23 (44.2%)	184 (45.4%)	
<i>Ethnicity (n, %)</i>				0.97
Caucasian North African	139 (39.4%)	20 (38.5%)	159 (39.2%)	
Caucasian European	188 (53.2%)	29 (55.8%)	217 (53.6%)	
Other	26 (7.4%)	3 (5.8%)	29 (7.2%)	
<i>Diabetes type (n, %)</i>				0.01
Type 1	113 (32.0%)	7 (13.5%)	120 (29.6%)	
Type 2	168 (47.6%)	34 (65.4%)	202 (49.9%)	
Other	30 (8.5%)	2 (3.8%)	32 (7.9%)	

Unknown	42 (11.9%)	9 (17.3%)	51 (12.6%)	
<i>Last eye exam (n, %)</i>				0.83
≤ 12 months	140 (39.7%)	24 (46.1%)	164 (40.5%)	
12-24 months	73 (20.7%)	10 (19.2%)	83 (20.5%)	
≥ 24 months	140 (39.7%)	18 (34.6%)	158 (39.0%)	
<i>Diabetes duration</i>				0.02
Median (years, range)	13 (0-56)	15 (1-45)	-	
Mean (years)	15	19	-	

Table I. Characteristics of patients included and excluded from the primary analysis. Other type of diabetes includes: Monogenic diabetes, diabetes secondary to chronic pancreatitis, corticotherapy-related diabetes, post-transplantation diabetes. Other ethnicity includes: Sub-Saharan African and Asian.

	Primary analysis (n=353)	Sensitivity analysis scenario (n=405)	
	Complete-case set (95% CI)	Best-case (95% CI)	Worst-case (95% CI)
Prevalence	15.3%	-	-
Sensitivity (%)	88.9 [77.4-95.8]	90.2 [79.8-96.3]	48.5 [38.3-58.7]
Specificity (%)	98.7 [96.7-99.7]	98.8 [97.0-99.7]	96.4 [93.6-98.2]
AUC (%)	96.5 [91.8-99.7]	-	-
Predictive value			
Positive (%)	92.3 [81.4-97.9]	93.2 [83.5-98.1]	81.3 [69.1-90.3]
Negative (%)	98.0 [95.7-99.3]	98.3 [96.3-99.4]	85.3 [81.1-88.8]

		AI grading			Total
		NR	RDR	Ungradable (excluded)	
Human grading	NR	295	4	0	299
	RDR	6	48	0	54
	Ungradable (excluded)	14	7	31	52
	Total	315	59	31	405

Table II.
Cross
tabulation
results
between AI
system and
human
grading.

Table III. Diagnostic accuracy for referable diabetic retinopathy. AUC= Area under the curve. CI= confidence interval. NR=Non Referable. RDR=Referable Diabetic Retinopathy. AI=Artificial intelligence.

Subgroup	Prevalence	Sensitivity % (95% CI)	Specificity % (95% CI)	AUC % (95% CI)	Comparison AUC p value
Age					
19-52 years	5.4%	84.2 [60.4-96.7]	99.1 [95.1-100]	99.8 [99.1-100]	Reference
53-64 years	5.4%	89.5 [66.9-98.7]	99.0 [94.7-100]	91.0 [77.2-100]	0.16
65-89 years	4.5%	93.7 [69.8-99.9]	97.6 [91.7-99.7]	98.7 [96.3-100]	0.23
Sex					
Female	6.2%	81.9 [59.7-94.9]	98.5 [94.9-99.9]	99.3 [98.2-100]	0.19
Male	9.1%	93.7 [79.2-99.2]	98.7 [95.5-99.9]	94.6 [86.5-100]	
Ethnicity					
European Caucasian	8.2%	86.2 [68.3-96.1]	99.3 [96.5-100]	96.5 [90-100]	0.82
North African Caucasian	5.1%	88.9 [65.3-98.6]	98.3 [94.1-99.8]	95.3 [85.2-100]	
Diabetes type					
Type 1	6.0%	81.0 [58.1-94.5]	99.0 [94.1-100]	99.2 [97.8-100]	Reference
Type 2	6.5%	95.6 [78.0-99.9]	98.0 [94.1-99.6]	96.2 [88.6-100]	0.39

Other*	2.8%	90 [55.5-99.7]	100 [94.22-100]	90.5 [70.48-100]	0.36
HbA1c					
<8%	4.8%	88.2 [63.5-98.5]	99.5 [97.2-100]	94.4 [83.2-100]	0.63
≥ 8%	10.2%	88.9 [74-96.9]	97.0 [91.6-99.4]	97.2 [92.5-99.9]	
Diabetes duration					
<10 years	4.2%	86.7 [59.5-98.3]	99.1 [95.4-100]	89.4 [72.5-100]	0.17
≥ 10 years	11.0%	89.7 [75.8-97.1]	98.3 [95.2-99.6]	99.4 [98.6-100]	
BMI					
<28 kg/m²	8.2%	86.2 [68.3-96.1]	98.7 [95.2-99.9]	96.1 [89.4-99.8]	0.90
≥ 28 kg/m²	7.4%	92.3 [74.0-99.0]	98.1 [94.6-99.6]	96.7 [89.7-100]	
Quality of image					
Excellent	5.7%	85.0 [62.1-96.8]	93.3 [94.1-99.8]	99.6 [98.7-100]	Reference
Good	8.0%	92.8 [76.5-99.1]	99.2 [95.8-100]	93.6 [83.8-100]	0.17
Adequate or dirty lens	1.7%	83.3 [35.8-99.5]	97.9 [88.7-100]	98.9 [96.4-100]	0.40

Table IV. Sub-group analysis (n = 353). AUC=area under the curve. DR=diabetic retinopathy. BMI=body mass index. HbA1c=glycated hemoglobin. CI=confidence interval. *Other type of diabetes includes: Monogenic diabetes, diabetes secondary to chronic pancreatitis, corticotherapy-related diabetes, post-transplantation diabetes.

	Artificial intelligence model		Human grader model	
	OR (95% CI)	P value	OR (95% CI)	P value
Age (per 1-year increase)	1.02 [1.00-1.04]	0.12	1.00 [0.98-1.02]	0.99
Ethnicity				

European Caucasian	Reference			
North African Caucasian (vs reference)	0.97 [0.50-1.84]	0.91	0.77 [0.38-1.52]	0.46
Other (vs reference)	2.96 [1.07-7.79]	0.03	2.34 [0.81-6.25]	0.1
Diabetes duration (per 1-year increase)	1.03 [1.00-1.06]	0.04	1.03 [1.00-1.06]	0.04
HbA1c (per 1% increase)	1.29 [1.13-1.47]	1.4×10⁻⁴	1.30 [1.15-1.49]	6.3×10⁻⁴
BMI (per 1 kg/m ² increase)	0.99 [0.94-1.04]	0.65	1.00 [0.95-1.06]	0.92

Table V. Multivariate analysis of systemic risk factors with referable diabetic retinopathy diagnosed by the ensemble AI model, as compared with human graders.

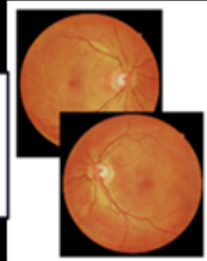
Patients were the units of analysis (n=353). BMI=body mass index.

HbA1c=glycated hemoglobin. OR=odds ratio. CI=confidence interval

Diabetes
consultation



Immediate DR screening
10' – 20'
n=405



Automated report for the patient
referable status
(cf Fig2)



RDR : Recommend to consult an
ophthalmologist for a follow up

Human grading



Included patients in the
primary analysis :
n=353 (87,2%)

Excluded Patients from the
primary analysis
n= 52 (12,8%)

- Small pupil (n=14, 26,9%)
- Poor fixation (n=1, 1,9%)
- Technical issues (n=8, 15,4%)

- Insufficient Quality on at least
one eye
(n=8, 15,4%, cf Fig3)

- Ungradable images
(n=21, 40,4%)

Patient contacted for an
appointment at the Erasme
Ophthalmology Department if :

- False negatives
- True positives without
regular follow-up with the
ophthalmologist
- If the patient requests it



NR:
n= 299 (84.7%)

RDR:
n=54 (15.3%)

"New" cases:
n=22 (40.7%)

Eye Screening with Artificial Intelligence Diabetic Retinopathy and Diabetic Maculopathy

Patient details

Name:
Medical identifier:
Birth date:
Screening date:
Workspace: HUB Erasme

Right eye



Left eye



Referable diabetic retinopathy detected!

Both values (R: 2.889; L: 2.414) are above our threshold of 1.371, the patient will be referred!

No refractable diabetic maculopathy detected!

Both values (R: 0.064; L: 0.008) are beneath our threshold of 0.38, the patient will not be referred for diabetic maculopathy.

Notes

Eye Screening with Artificial Intelligence Diabetic Retinopathy and Diabetic Maculopathy

Patient details

Name:
Medical identifier:
Birth date:
Screening date:
Workspace: HUB Erasme

Right eye



Left eye



No refractable diabetic retinopathy detected!

Both values (R: 0.189; L: 0.811) are beneath our threshold of 1.371, the patient will not be referred for diabetic retinopathy.

No refractable diabetic maculopathy detected!

Both values (R: 0.000; L: 0.006) are beneath our threshold of 0.38, the patient will not be referred for diabetic maculopathy.

Notes

